



Shore School

2014
Year 12 Mid-Year
Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/or calculations
- Write your examination number on the front cover of each booklet
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Examination Number:

Set:

Total marks – 70

Section I Pages 2–5

10 marks

- Attempt questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–10

60 marks

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section

Note: Any time you have remaining should be spent revising your answers.

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

Section I

10 marks

Attempt Questions 1–10

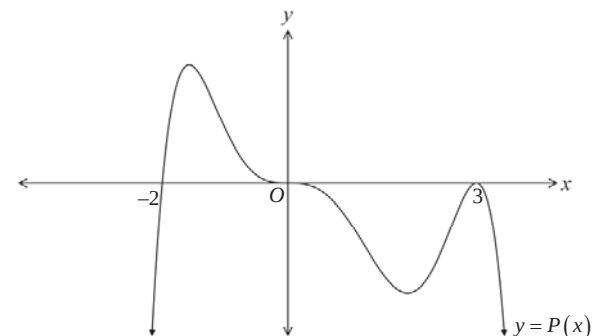
Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet for Questions 1–10.

- 1 Which expression is a correct factorisation of $27x^3 + 8$?

- (A) $(3x + 2)(9x^2 - 12x + 4)$
(B) $(3x + 2)(9x^2 - 6x + 4)$
(C) $(3x + 2)(9x^2 + 12x + 4)$
(D) $(3x + 2)(9x^2 + 6x + 4)$

- 2 A polynomial is sketched on the axes below.



Which of the following could be the equation of the polynomial?

- (A) $P(x) = -x^3(x + 2)(x - 3)^2$
(B) $P(x) = x(x + 2)(x - 3)$
(C) $P(x) = x^3(x + 2)(x - 3)^2$
(D) $P(x) = x^3(x - 2)(x + 3)^2$

3 What is $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{4}}{2x}$?

(A) 0

(B) $\frac{1}{8}$

(C) $\frac{1}{2}$

(D) 8

4 What is the derivative of $\sin^{-1}(5x)$?

(A) $\frac{-5}{\sqrt{1-25x^2}}$

(B) $\frac{-1}{\sqrt{1-25x^2}}$

(C) $\frac{1}{\sqrt{1-25x^2}}$

(D) $\frac{5}{\sqrt{1-25x^2}}$

5 Given that $\sin \theta = \frac{2t}{1+t^2}$ and $\cos \theta = \frac{1-t^2}{1+t^2}$ when $t = \tan \frac{\theta}{2}$, which of the following is a simplified expression for $\sin \theta \sec \theta$?

(A) $\frac{2}{1-t}$

(B) $\frac{1-t}{2}$

(C) $\frac{1-t^2}{2t}$

(D) $\frac{2t}{1-t^2}$

6 Consider the function $f(x) = \frac{3x^4 + 5x^2}{x^4 + 5}$.

Which one of the following statements is correct?

(A) $f(x)$ is odd and $\lim_{x \rightarrow \infty} f(x) = 3$

(B) $f(x)$ is odd and $\lim_{x \rightarrow \infty} f(x) = 5$

(C) $f(x)$ is even and $\lim_{x \rightarrow \infty} f(x) = 3$

(D) $f(x)$ is even and $\lim_{x \rightarrow \infty} f(x) = 5$

7 The cubic polynomial $P(x)$ has roots α , β , and γ such that $\alpha + \beta + \gamma = 5$ and $\alpha\beta + \alpha\gamma + \beta\gamma = 4$

Which of the following could be the equation of $P(x)$?

(A) $P(x) = x^3 + 5x^2 - 4x - 20$

(B) $P(x) = x^4 - 5x^3 + 4x^2 - x + 11$

(C) $P(x) = 2x^3 - 5x^2 + 4x - 20$

(D) $P(x) = 2x^3 - 10x^2 + 8x + 17$

8 Which of the following gives all of the solutions to $\left(1 - \frac{1}{x}\right)^2 + 3\left(1 - \frac{1}{x}\right) - 4 = 0$?

(A) $x = -4$, $x = 1$

(B) $x = 4$, $x = -1$

(C) $x = \frac{1}{5}$ only

(D) $x = \frac{1}{5}$, $x = 0$

9 What is the derivative of $x \sec^2 x$?

(A) $\frac{\cos^2 x + x \sin 2x}{\cos^4 x}$

(B) $\frac{\cos^2 x - 2x \cos x}{\cos^4 x}$

(C) $\frac{-2x}{\cos^3 x}$

(D) $x \tan x + \sec^2 x$

10 It is known that $\tan A = 4$ and $\tan(A + B) = \frac{1}{2}$.

What is the value of $\tan B$?

(A) $-3\frac{1}{2}$

(B) $-\frac{7}{6}$

(C) -1.292

(D) -0.862

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) Solve the inequality $\frac{x+3}{x-2} > -1$. 3

(b) Consider the function $f(x) = e^{5x+4}$. 2
Find $f^{-1}(x)$, the inverse function of $f(x)$.

(c) Find $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$, giving your answer in terms of π . 2

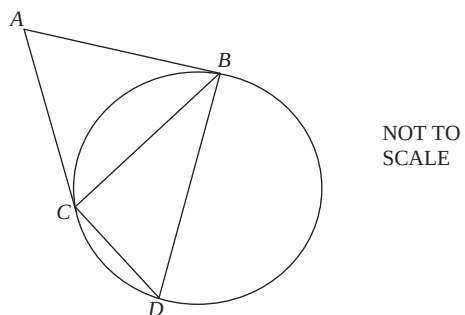
(d) Using the substitution $u = 1 + 2x$, find $\int \frac{6 dx}{\sqrt{(1+2x)^3}}$. 3

(e) The acute angle between the lines $3x - y + 7 = 0$ and $mx - y + 1 = 0$ is 45° . 2
Find the possible value(s) of m .

(f) When the polynomial $P(x) = x^3 + px^2 + qx - 4$ is divided by $(x - 2)$ the remainder is 12. Also, $(x + 1)$ is a factor of $P(x)$. 3
Find the values of p and q .

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Express $3\cos x + \sqrt{3}\sin x$ in the form $R\cos(x - \alpha)$, where $0 < \alpha < \frac{\pi}{2}$ and $R > 0$. 2
- (ii) Hence, or otherwise, find the general solution of the equation $3\cos x + \sqrt{3}\sin x = \sqrt{6}$. 3
- (b) AB and AC are tangents to a circle. D is a point on the circle such that $2 \times \angle DBC = \angle BAC$ and $\angle BDC = \angle BAC$.



Copy or trace the diagram into your writing booklet.

- (i) Show that DB is a diameter. 3
- (ii) Show that $BC = AB$. 1

Question 12 continues on page 8

Question 12 (continued)

- (c) Consider the function $f(x) = 4x - x^3$.
- (i) Sketch $y = f(x)$, showing the x and y intercepts and the coordinates of the stationary points. 3
- (ii) Find the largest domain containing the origin for which $f(x)$ has an inverse function, $f^{-1}(x)$. 1
- (iii) State the domain of $f^{-1}(x)$. 1
- (iv) Find the gradient of the inverse function at the origin. 1

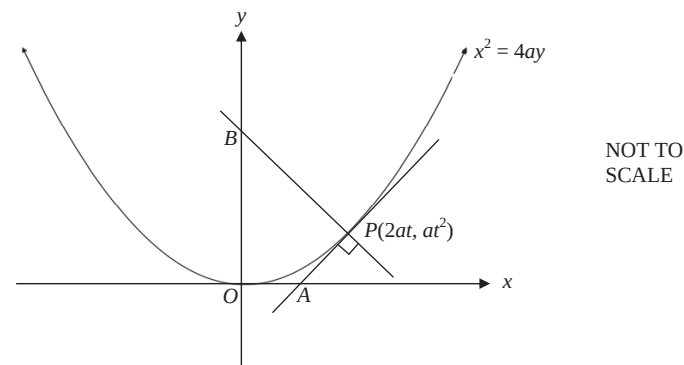
End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Draw a neat graph of $y = 3\sin 2x$, for $-\pi \leq x \leq \pi$, indicating all intercepts on the x axis. 2
- (ii) Use your graph to determine the number of solutions to the equation $3\sin 2x = x$. 1
- (iii) If $m > 0$, for what values of m will the equation $3\sin 2x = mx$ have only one solution? 2
- (b) Prove by mathematical induction that for all integers $n \geq 2$, 3
- $$1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + (n-1) \times n = \frac{1}{3}(n-1)n(n+1).$$
- (c) Consider the function $f(x) = x \log_e x - 1$ with $x > 0$.
- (i) Find the coordinates of the stationary point on $y = f(x)$ and determine its nature. 2
- (ii) Let $x = 2$ be a first approximation to the root of the equation $x \log_e x - 1 = 0$. 2
- Use one application of Newton's method to approximate the x -intercept. Leave your answer correct to 2 decimal places.
- (iii) Explain why the curve $y = f(x)$ is concave up for all $x > 0$. 1
- (iv) Sketch the curve, showing all its main features. 2

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) The point $P(2at, at^2)$ lies on the parabola $x^2 = 4ay$.



- (i) Show that the tangent to the parabola at P meets the x axis at $A(at, 0)$. 2
- (ii) Show that the normal to the parabola at P meets the y axis at $B(0, 2a + at^2)$. 2
- (iii) The point R divides BA **externally** in the ratio $2:1$.
- (α) Show that the coordinates of R are $(2at, -2a - at^2)$. 1
- (β) Show that R lies on a parabola with the same directrix and focal length as the original parabola. 3

- (b) Consider the function $y = \frac{1}{2} \cos^{-1}(x-1)$.

- (i) What is the domain and range of the function? 2
- (ii) Sketch the graph of the function showing the coordinates of the endpoints. 1
- (iii) The region in the first quadrant bounded by the curve $y = \frac{1}{2} \cos^{-1}(x-1)$ and the coordinate axes is rotated about the y -axis. Find the volume of the solid of revolution. Express your answer in simplest exact form. 4

End of paper

Year 12 Mid-Year Extension 1 Maths Solutions

Section I

1. B
2. A
3. B

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin \frac{x}{4}}{2x} &= \lim_{x \rightarrow 0} \frac{\sin \frac{x}{4}}{\frac{x}{4}} \cdot \frac{1}{8} \\ &= 1 \times \frac{1}{8} \\ &= \frac{1}{8}.\end{aligned}$$

$$\begin{aligned}4. D. \quad \frac{d(\sin^{-1}(5x))}{dx} &= \frac{1}{\sqrt{1-(5x)^2}} \cdot 5 \\ &= \frac{5}{\sqrt{1-25x^2}}\end{aligned}$$

$$\begin{aligned}5. D. \quad \sin \theta \sec \theta &= \sin \theta \cdot \frac{1}{\cos \theta} \\ &= \frac{\sin \theta}{\cos \theta} \\ &= \frac{2t}{1+t^2} \cdot \frac{1-t^2}{1+t^2} \\ &= \frac{2t}{1-t^2}\end{aligned}$$

$$\begin{aligned}6. C. \quad \lim_{x \rightarrow \infty} \frac{3x^4 + 5x^2}{x^4 + 5} &= \lim_{x \rightarrow \infty} \frac{3 + \frac{5}{x^2}}{1 + \frac{5}{x^4}} \\ &= 3.\end{aligned}$$

Section I (cont.)

$$\begin{aligned}7. D. \quad \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} \\ &= 4\end{aligned}$$

$$\begin{aligned}\alpha + \beta + \gamma &= \frac{-b}{a} \\ &= 5.\end{aligned}$$

$$\begin{aligned}8. C \quad \text{Let } u &= 1 - \frac{1}{x} \\ u^2 + 3u - 4 &= 0 \\ (u+4)(u-1) &= 0 \\ u &= -4 \text{ or } u = 1 \\ \therefore 1 - \frac{1}{x} &= 1 \text{ or } 1 - \frac{1}{x} = -4 \\ -\frac{1}{x} &= 0 \quad \quad \quad -\frac{1}{x} = -5 \\ \text{no solution} \quad \quad \quad x &= \frac{1}{5} \\ \therefore x &= \frac{1}{5} \text{ only.}\end{aligned}$$

$$\begin{aligned}9. A \quad \frac{d(x \cdot \sec^2 x)}{dx} &= \frac{d}{dx} \left(\frac{x}{\cos^2 x} \right) \\ &= \frac{1 \cdot \cos^2 x - 2 \cdot \cos x \cdot \sin x \cdot x}{(\cos^2 x)^2} \\ &= \frac{\cos^2 x + x \cdot 2 \sin x \cos x}{\cos^4 x} \\ &= \frac{\cos^2 x + x \cdot \sin 2x}{\cos^4 x}.\end{aligned}$$

Section I (cont.)

10. B. $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$$\frac{1}{2} = \frac{4 + \tan B}{1 - 4 \tan B}$$

$$1 - 4 \tan B = 8 + 2 \tan B$$

$$-7 = 6 \tan B$$

$$\tan B = \frac{-7}{6}$$

Section 2

Q11

a) $\frac{x+3}{x-2} > -1, \quad x \neq 2.$

Let $x+3 = -(x-2)$

$$2x = -\frac{1}{2} - 1$$

$$x = -\frac{1}{2}$$

Test $x=0$

$$\frac{0+3}{0-2} = -\frac{3}{2}$$

$$\nless -1$$

$\therefore$



$$x > 2 \text{ or } x < -\frac{1}{2}$$

OR

11a) $\frac{x+3}{x-2} > -1, \quad x \neq 2.$

$$(x+3)(x-2) > -(x-2)^2$$

$$x^2 + x - 6 > -x^2 + 4x - 4$$

$$2x^2 - 3x - 2 > 0$$

Let $2x^2 - 3x - 2 = 0$

$$(2x+1)(x-2) = 0$$

$$x = 2 \text{ or } x = -\frac{1}{2}$$

Test $x=0$. (as above).

$$\therefore x > 2 \text{ or } x < -\frac{1}{2}$$

b) $f(x) = e^{5x+4}$

Let $y = e^{5x+4}$

$$\ln y = 5x+4$$

$$x = \frac{\ln y - 4}{5}$$

$$\therefore f^{-1}(x) = \frac{\ln x - 4}{5}$$

c) $\int_0^1 \frac{dx}{\sqrt{4-x^2}} = \int_0^1 \frac{dx}{\sqrt{2^2-x^2}}$

$$= \left[\sin^{-1}\left(\frac{x}{2}\right) \right]_0^1$$

$$= \sin^{-1}\left(\frac{1}{2}\right) - \sin^{-1}(0)$$

$$= \frac{\pi}{6}$$

$$\begin{aligned}
 11 d) \quad & \int \frac{6dx}{\sqrt{(1+2x)^3}} \\
 &= \int \frac{3du}{\sqrt{u^3}} \\
 &= 3 \int u^{-\frac{3}{2}} du \\
 &= 3 \frac{u^{-\frac{1}{2}}}{-\frac{1}{2}} + C \\
 &= \frac{-6}{\sqrt{u}} + C \\
 &= \frac{-6}{\sqrt{1+2x}} + C.
 \end{aligned}$$

$$\begin{aligned}
 \text{Let } u &= 1+2x \\
 \frac{du}{dx} &= 2 \\
 du &= 2dx \\
 3du &= 6dx
 \end{aligned}$$

$$\begin{aligned}
 e) \quad \tan \alpha &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\
 1 &= \left| \frac{3 - m}{1 + 3m} \right|
 \end{aligned}$$

$$\begin{aligned}
 m_1 &= 3 \\
 m_2 &= m
 \end{aligned}$$

$$\begin{aligned}
 1+3m &= 3-m & \text{or} \\
 4m &= 2 & \text{or} \\
 m &= \frac{1}{2} & \text{or}
 \end{aligned}$$

$$\begin{aligned}
 1+3m &= m-3 \\
 2m &= -4 \\
 m &= -2.
 \end{aligned}$$

$$\begin{aligned}
 Q11 f) \quad & P(2) = 12 \\
 & 2^3 + p \cdot 2^2 + q \cdot 2 - 4 = 12 \\
 & 8 + 4p + 2q - 4 = 12 \\
 & 4p + 2q = 8 \quad (1)
 \end{aligned}$$

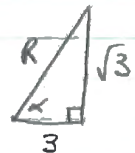
$$\begin{aligned}
 & P(-1) = 0 \\
 & (-1)^3 + p(-1)^2 + q(-1) - 4 = 0 \\
 & -1 + p - q - 4 = 0 \\
 & p - q = 5 \quad (2) \\
 & p = 5 + q \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 (3) \rightarrow (1): \quad & 4(5+q) + 2q = 8 \\
 & 20 + 4q + 2q = 8 \\
 & 6q = -12 \\
 & q = -2 \\
 \therefore p &= 3, q = -2.
 \end{aligned}$$

Q12

$$\begin{aligned}
 a) i) \quad & R \cos(x - \alpha) = R \cos x \cos \alpha + R \sin x \sin \alpha \\
 \text{So } & 3 \cos x + \sqrt{3} \sin x = R \cos x \cos \alpha + R \sin x \sin \alpha \\
 \text{Equating coefficients}
 \end{aligned}$$

$$R \cos \alpha = 3, \quad R \sin \alpha = \sqrt{3}.$$



$$\text{So } \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6} \quad \text{for } 0 < \alpha < \frac{\pi}{2}$$

$$R^2 = 3^2 + (\sqrt{3})^2$$

$$R = \sqrt{12}, \quad R > 0.$$

$$\therefore 3 \cos x + \sqrt{3} \sin x = \sqrt{12} \cos(x - \frac{\pi}{6}).$$

Q12a) ii)

$$3\cos x + \sqrt{3}\sin x = \sqrt{6}$$

$$\text{So } \sqrt{12} \cos(x - \frac{\pi}{6}) = \sqrt{6}$$

$$\cos(x - \frac{\pi}{6}) = \sqrt{\frac{1}{2}}$$

$$\cos(x - \frac{\pi}{6}) = \frac{1}{\sqrt{2}}$$

$$x - \frac{\pi}{6} = \pm \frac{\pi}{4} + 2n\pi, \text{ where } n \text{ is an integer}$$

$$\therefore x = \frac{\pi}{6} + \frac{\pi}{4} + 2n\pi \text{ or } x = \frac{\pi}{6} - \frac{\pi}{4} + 2n\pi$$

$$= \frac{5\pi}{12} + 2n\pi \text{ or } x = -\frac{\pi}{12} + 2n\pi, \quad n \in \mathbb{Z}$$

b) $\angle ABC = \angle BDC$ (alternate segment theorem)

$\angle ACB = \angle BDC$ (alternate segment theorem)

$\angle BAC = \angle BDC$ (given)

$\therefore \triangle ABC$ is equilateral and

$$\angle ABC = \angle ACB = \angle BAC = \angle BDC = 60^\circ$$

$$\angle DBC = \angle BAC \div 2 \quad (\text{given})$$

$$= 30^\circ$$

Now $\angle ABD = \angle ABC + \angle DBC$

$$= 60^\circ + 30^\circ$$

$$= 90^\circ$$

$\therefore DB$ is a diameter
(tangent and diameter through point of contact meet at right angles).

Q12b) ii)

From (i), $\triangle BAC$ is equilateral.
 $\therefore BC = AB$.

c) (i) $f(x) = 4x - x^3$

$$= x(4 - x^2)$$

$$= x(2 - x)(2 + x)$$

y-intercept when $x = 0$

$$y = 0 - 0 = 0$$

x-intercept when $y = 0$

$$0 = x(2 - x)(2 + x)$$

$$x = 0, \pm 2$$

$$f'(x) = 4 - 3x^2$$

Stat. pts when $f'(x) = 0$

$$0 = 4 - 3x^2$$

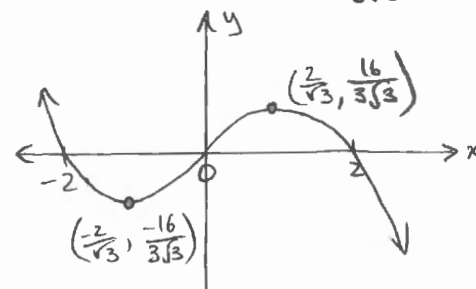
$$x^2 = \frac{4}{3}$$

$$x = \pm \frac{2}{\sqrt{3}}$$

When $x = \frac{2}{\sqrt{3}}, f(x) = 4\left(\frac{2}{\sqrt{3}}\right) - \left(\frac{2}{\sqrt{3}}\right)^3$

$$= \frac{8}{\sqrt{3}} - \frac{8}{3\sqrt{3}} \text{ or } \frac{-8}{\sqrt{3}} + \frac{8}{3\sqrt{3}}$$

$$= \frac{16}{3\sqrt{3}} \text{ or } -\frac{16}{3\sqrt{3}}$$



Q12 c) ii)

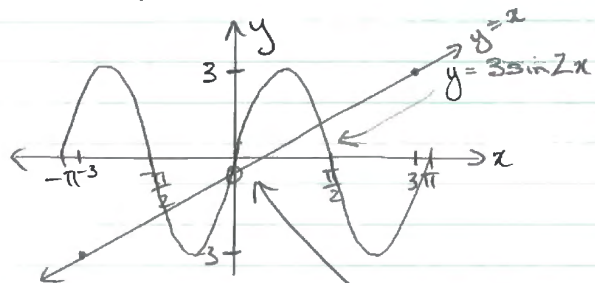
$f(x)$ is monotonic increasing for
 $-\frac{2}{\sqrt{3}} \leq x \leq \frac{2}{\sqrt{3}}$.

$\therefore$ this is the largest domain containing the origin for which $f^{-1}(x)$ exists.

iii) Domain of $f^{-1}(x)$: $-\frac{16}{3\sqrt{3}} \leq x \leq \frac{16}{3\sqrt{3}}$

iv) $f'(x) = 4 - 3x^2$
 $f'(0) = 4$
 $\therefore$ gradient of inverse function at origin is $\frac{1}{4}$.

Q13 a) i) Period = $\frac{2\pi}{2}$
 $= \pi$
 Amplitude = 3.



ii) see above
 $\therefore$ 3 solutions.

Q13 a)

iii) $3 \sin 2x = mx$ will have only one solution when

$$\frac{d(mx)}{dx} \geq \frac{d(3 \sin 2x)}{dx} \Big|_{x=0}.$$

$$m \geq 6 \cos 2x \Big|_{x=0}.$$

$$m \geq 6.$$

b) Prove true when $n=2$.

$$\begin{aligned} \text{LHS} &= 1 \times 2 \\ &= 2 \end{aligned} \quad \begin{aligned} \text{RHS} &= \frac{1}{3} \times 1 \times 2 \times 3 \\ &= \frac{1}{3} \times 6 \\ &= 2 \end{aligned}$$

LHS = RHS
 $\therefore$ true for $n=2$.

Assume true for $n=k$

i.e.

$$1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + (k-1)k = \frac{1}{3}(k-1)k(k+1).$$

Prove true for $n=k+1$.

i.e. required to prove that

$$1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + (k-1)k + k(k+1) = \frac{1}{3}k(k+1)(k+2).$$

$$\begin{aligned} \text{LHS} &= \frac{1}{3}(k-1)k(k+1) + k(k+1) \quad \text{by the inductive hypothesis} \\ &= \frac{1}{3}k(k+1)[(k-1) + 3] \\ &= \frac{1}{3}k(k+1)(k+2) \\ &= \text{RHS.} \end{aligned}$$

Q13) b)

$\therefore$ by the principle of mathematical induction, statement is true for $n \geq 2, n \in \mathbb{Z}$.

c) i) $f(x) = x \ln x - 1, x > 0$

$$f'(x) = 1 \cdot \ln x + \frac{1}{x} \cdot x - 0$$

$$= 1 + \ln x$$

Stat. pts when $f'(x) = 0$

$$0 = 1 + \ln x$$

$$\ln x = -1$$

$$x = e^{-1}$$

$$= \frac{1}{e}$$

when $x = \frac{1}{e}, f(x) = \frac{1}{e} \cdot -1 - 1$

$$= -\frac{1}{e} - 1$$

$$f''(x) = \frac{1}{x}$$

$$> 0 \text{ when } x > 0.$$

$\therefore$ stationary point is a ^{local} minimum.
(curve is concave up).

$\therefore$ the stationary point at $(\frac{1}{e}, -\frac{1}{e} - 1)$ is a minimum turning point.

Q13) c) ii)

$$f(x) = x \ln x - 1$$

$$f'(x) = \ln x + 1$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 2 - \frac{2 \cdot \ln 2 - 1}{\ln 2 + 1}$$

$$= 1.771848327 \dots$$

$$\hat{=} 1.77$$

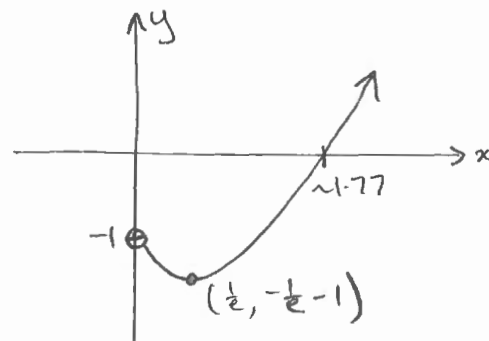
$\therefore$ the x -intercept is approximately $x \hat{=} 1.77$

iii) $f''(x) = \frac{1}{x}$ and $x > 0$

$$\therefore f''(x) > 0 \text{ for } x > 0$$

$\therefore y = f(x)$ is concave up for $x > 0$.

iv)



$$\lim_{x \rightarrow 0} x \ln x - 1 = -1$$

Q14 a)

i) gradient of tangent at $P = t$

tangent at P : $y - at^2 = t(x - 2at)$

$$y = tx - at^2$$

When $y = 0$:

$$0 = tx - at^2$$

$$at^2 = tx$$

$$x = at$$

$\therefore A = (at, 0)$.

ii) gradient of normal at $P = -\frac{1}{t}$

normal at P : $y - at^2 = -\frac{1}{t}(x - 2at)$

$$yt - at^3 = -x + 2at$$

When $x = 0$:

$$yt - at^3 = 2at$$

$$yt = 2at + at^3$$

$$y = 2a + at^2$$

$\therefore B = (0, 2a + at^2)$.

iii) $\propto$ $A(at, 0)$ $B(0, 2a + at^2)$

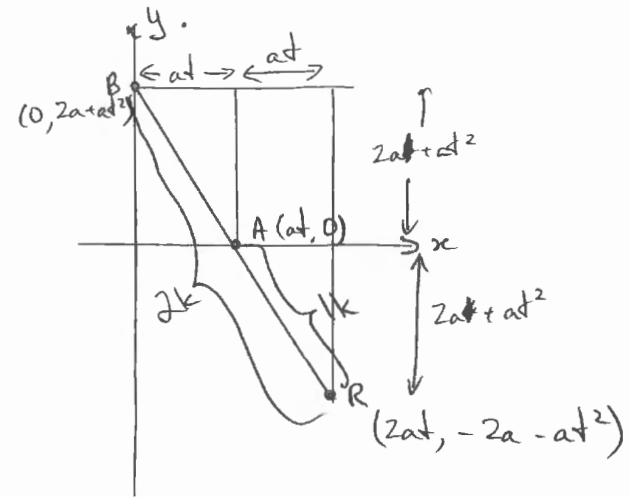
$$-1:2$$

$\therefore x = 2at, y = -2a - at^2$

$\therefore R = (2at, -2a - at^2)$.

OR

$\propto$ cont.



By similar triangles,

$$R = (2at, -2a - at^2)$$

β) $x = 2at, y = -2a - at^2$
 $t = \frac{x}{2a}$

$$\therefore y = -2a - a\left(\frac{x}{2a}\right)^2$$

$$= -2a - \frac{ax^2}{4a^2}$$

$$= -2a - \frac{x^2}{4a}$$

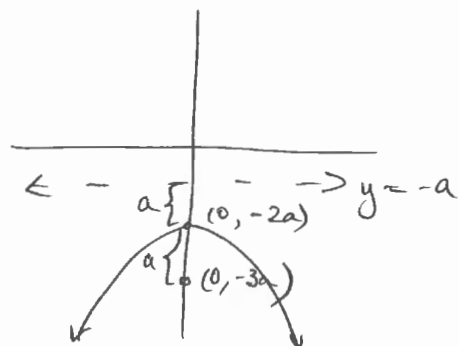
$$\frac{x^2}{4a} = -y - 2a$$

$$= -(y + 2a)$$

$$x^2 = -4a(y + 2a)$$

14 a) iii) β) cont.

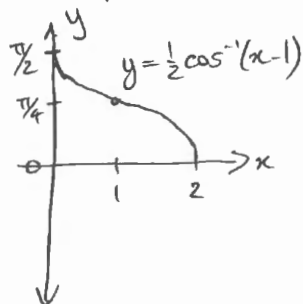
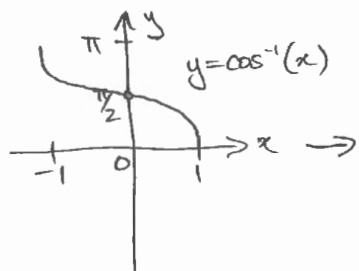
So ~~vertex~~ this is a concave down parabola with focal length $= a$, vertex $= (0, -2a)$



$\therefore$ directrix is $y = -a$.

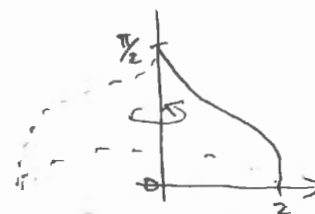
$\therefore$ focal length $= a$ and directrix ~~is~~ $y = -a$, same as original parabola.

14) b) ii)



i) Domain: $0 \leq x \leq 2$
Range: $0 \leq y \leq \frac{\pi}{2}$

Q14) b) iii)



$$\begin{aligned} y &= \frac{1}{2} \cos^{-1}(x-1) \\ 2y &= \cos^{-1}(x-1) \\ x-1 &= \cos 2y \\ x &= \cos 2y + 1 \end{aligned}$$

$$V = \pi \int_0^{\pi/2} x^2 dy.$$

$$= \pi \int_0^{\pi/2} (\cos 2y + 1)^2 dy.$$

$$= \pi \int_0^{\pi/2} \cos^2 2y + 2\cos 2y + 1 dy.$$

$$= \pi \int_0^{\pi/2} \frac{\cos 4y + 1}{2} + 2\cos 2y + 1 dy.$$

$$= \pi \left[\frac{\sin 4y}{8} + \frac{3y}{2} + \sin 2y \right]_0^{\pi/2}$$

$$= \pi \left[\left(\frac{\sin 2\pi}{8} + \frac{3\pi}{4} + \sin \pi \right) - (0 + 0 + 0) \right]$$

$$= \pi \cdot \frac{3\pi}{4}$$

$$= \frac{3\pi^2}{4} \text{ units}^3$$